# Experiment: 6
# 5G NR Frame Structure Analysis with Different Subcarrier Spacing

**Aim**

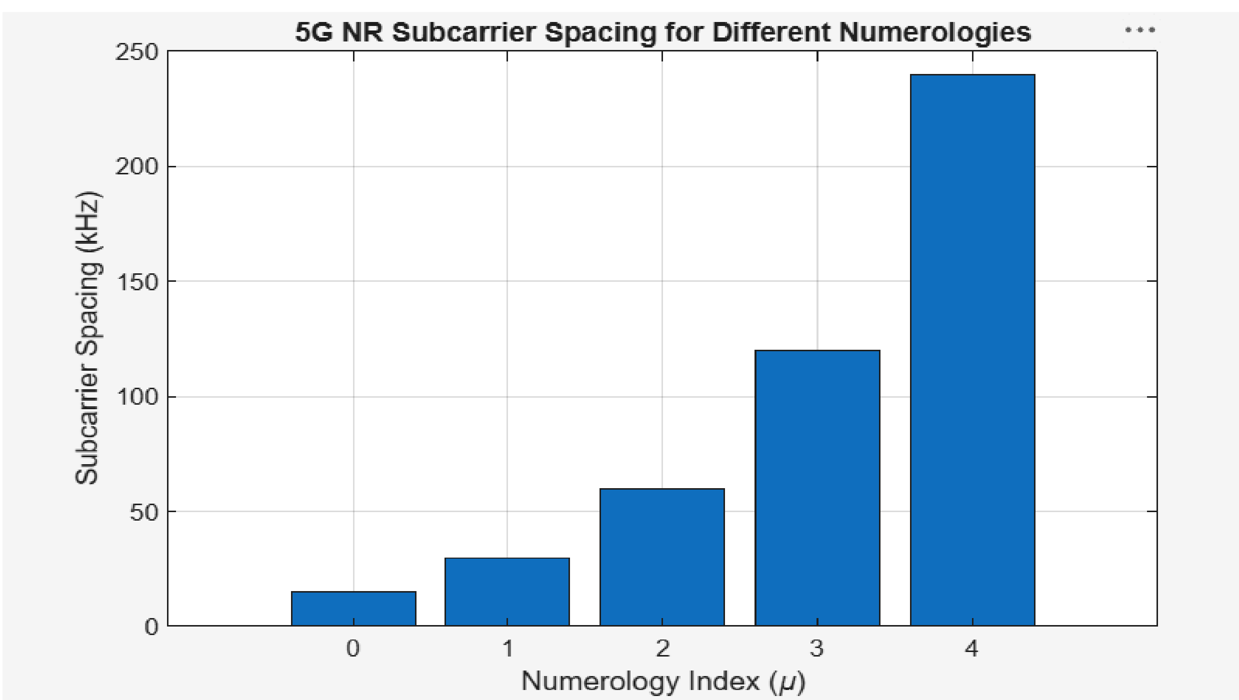
To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

**Objectives**

1. To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.
2. To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.
3. To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

**Objective – 1: Analyze Different Subcarrier Spacing Values**

```
clc;
clear;
close all;
scs = [15 30 60 120 240];
mu = 0:4;
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
% Subcarrier Spacing (kHz)
% Numerology Index
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```



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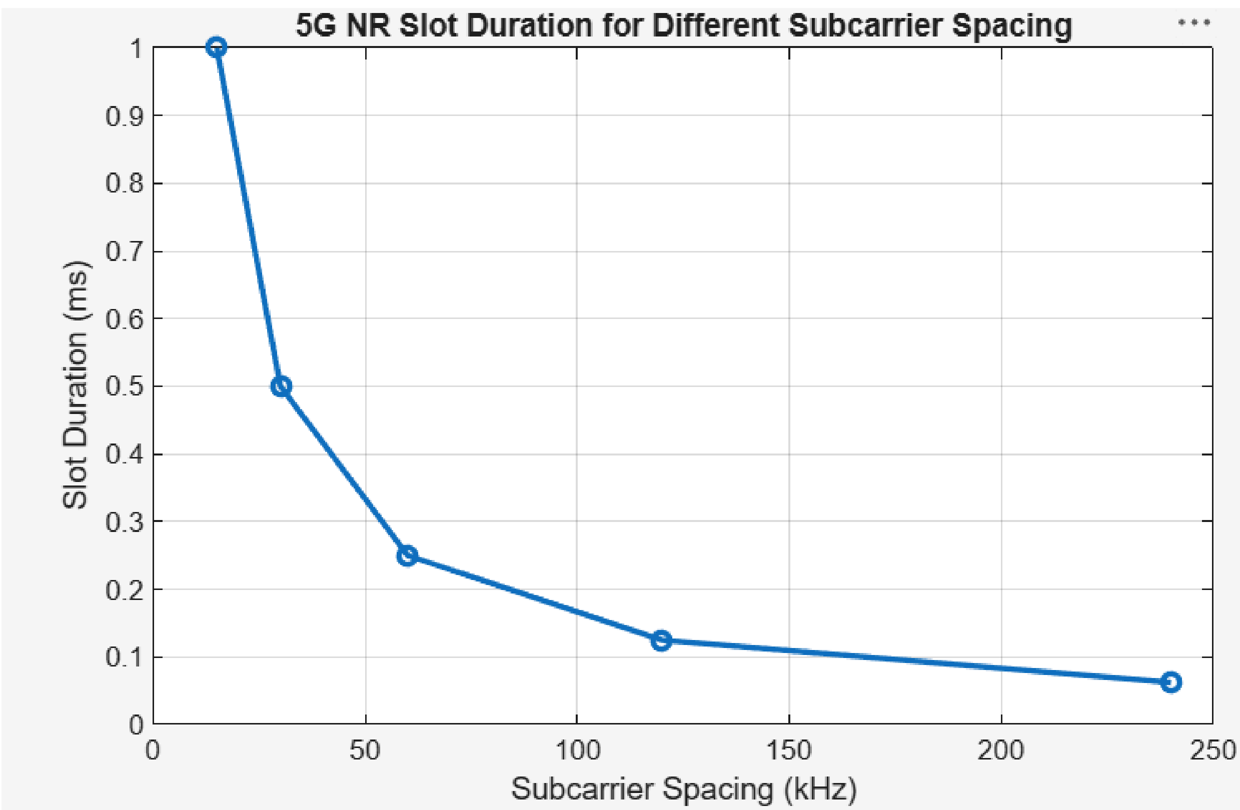
**5G NR Frame Structure Analysis with Different Subcarrier Spacing**

**Objective – 2 Compare Slot Duration**

```
clear;
clc;
close all;
scs = [15 30 60 120 240];
% Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625];  % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```



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**Objective – 3 Study the Number of OFDM Symbols**

```
clc;
clear;
close all;
scs = [15 30 60 120 240];
symbols = [14 14 14 14 14];
figure
bar(scs, symbols)
% Subcarrier Spacing (kHz)
% OFDM Symbols per Slot
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```